

lising some stock quotes and basic conversational patterns to make a full sentence. These keywords are ranked by importance, so that if a sentence contains two or more keywords, then ELIZA won't simply respond to the first keyword it comes across when reading the sentence from left to right. It will intelligently rank the keywords and respond accordingly.

Weizenbaum describes the need for the keyword approach and why it is successful by further elaborating on this “foreigner” analogy:

Consider the sentence “I am very unhappy these days”. Suppose a foreigner with only a limited knowledge of English but with a very good ear heard that sentence spoken but understood only the first two words, i.e. “I am”. Wishing to appear interested, perhaps even sympathetic, he may reply “How long have you been very unhappy these days?” What he must have done is apply a kind of template to the original sentence, one part of which matched the two words “I am” and the remainder isolated the words “very unhappy these days”. He must also have a reassembly kit specifically associated with that template, one that specifies that any sentence of the form “I am BLAH” can be transformed to “How long have you been BLAH”, independently of the meaning of BLAH. A somewhat more complicated example is given by the sentence “It seems that you hate me”. Here the foreigner understands only the words “you” and “me”; i.e., he applies a template that decomposes the sentence into the four parts – “It seems that” | “you” | “hate” | “me” – of which only the second (you) and fourth (me) parts are understood. The reassembly rule might then be “What makes you think I hate you”; i.e., it might throw away the first component, translate the two known words (“you” to “I” and “me” to “you”) and tack on a stock phrase (What makes you think) to the front of the reconstruction.

Given this basis for deconstructing language based on keywords, the next problem Weizenbaum had to tackle was suspension of disbelief. While talking to ELIZA, there needed to be a plausible scenario to accommodate the chatbot's obvious lack of being able to respond to every sentence intelligently. When it came up with seemingly repetitive responses to sentences, the other party in the conversation shouldn't have been able to figure out that it's a robot and not a person.

Weizenbaum decided to use the role of a psychiatrist. This mode of conversation was chosen because the psychiatric interview is one of the few examples of natural language communication in which one of the participating pair is free to assume the pose of knowing almost nothing of the real world. If, for example, one were to tell a psychiatrist “I went for